

MATH 3115 OMS & OED COMPLEX ANALYSIS

Lecture 4: Complex Integration

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Lesson Outcome

By the end of this lesson you should be able to:

- Define complex integrals .
- Apply De-Moivre theorem in evaluating complex integrals.
- Describe contour integrals
- Solve problems relating to contour integrals

Complex Integrals

Definition 1

If u and v are continuous functions on the interval $a \leq t \leq b$, then u and v are integral functions of t . Therefore, we have a definite integral of f

$$\int_a^b f(t)dt = \int_a^b u(t)dt + i \int_a^b v(t)dt$$

Complex Integrals

- By definition 1, We find anti-derivatives of u and v
- And evaluating the definite integrals on the right of the equation.
- That is $u'(t) = u(t)$ and $v'(t) = v(t)$ so that

$$\int_a^b f(t)dt = u(b) - u(a) + i(v(b) - v(a))$$

CONT'D

Example 2

Evaluate $\int_0^1 (t - i)^3 dt$ using binomial expansion.

Solution 3

$$\begin{aligned}\int_0^1 (t - i)^3 dt &= \int_0^1 \left(\binom{3}{0} t^3 (-i)^0 + \binom{3}{1} t^2 (-i)^1 + \binom{3}{2} t^1 (-i)^2 + \binom{3}{3} t^0 (-i)^3 \right) dt \\&= \int_0^1 (t^3 + 3t^2(-i) + 3t(1) + (-i)) dt \\&= \int_0^1 (t^3 - 3it^2 - 3t + i) dt \\&= \int_0^1 (t^3 - 3t) + i \int_0^1 (-3t^2 + 1) dt\end{aligned}$$

Solution 3 to example 2 CONT'D

$$\begin{aligned} &= \left. \frac{t^4}{4} - \frac{3}{2}t^2 \right|_0^1 + i \left. (-t^3 + t) \right|_0^1 \\ &= \left(\frac{1^4}{4} - \frac{3}{2} \right) + i(-1 + 1) \\ &= \left(\frac{1}{4} - \frac{3}{2} \right) + i(0) \\ &= \frac{1}{4} - \frac{3}{2} \\ &= \frac{1}{4} - \frac{6}{4} \\ &= -\frac{5}{4} \end{aligned}$$

complex Integral

Example 4

Evaluate $\int_0^1 (3t + it) dt$

De Moivre CONT'D

Solution 5

$$\begin{aligned}\int_0^1 (3t + it)dt &= \int_0^1 3tdt + i \int_0^1 tdt \\&= \frac{3}{2}t^2 \Big|_0^1 + i \frac{t^2}{2} \Big|_0^1 \\&= \frac{3}{2} + i(1/2) \\&= \frac{3}{2} + \frac{i}{2}\end{aligned}$$

Properties of Complex integrals

Definition

Let $f(t) = u(t) + iv(t)$ and $g(t) = p(t) + iq(t)$ be continuous functions on the interval $a \leq t \leq b$. Then we can write the following:

- (a) $\int_a^b (f(t) \pm g(t))dt = \int_a^b f(t)dt \pm \int_a^b g(t)dt$
- (b) $\int_a^b f(t)dt = \int_a^c f(t)dt + \int_c^b f(t)dt$ where the interval $a \leq t \leq b$ is submitted into two parts $a \leq t \leq c$ and $c \leq t \leq b$.
- (c) $\int_a^b (c + id)f(t) = (c + id) \int_a^b f(t)dt.$
- (k) $\int_b^a f(t)dt = - \int_a^b f(t)dt$
- (e) $\int_a^b f(t)dt = F(b) - F(a)$ where $F'(t) = f(t)$

Example 6

Show that $\int_0^{\pi} e^{it} dt = 2i$

Solution to example 6

Since the identity $e^{it} = \cos t + i \sin t$ (by Euler formula for complex exponential of trig function)

$$\begin{aligned}\int_0^\pi e^{it} dt &= \int_0^\pi (\cos t + i \sin t) dt \\ &= \int_0^\pi \cos t dt + i \int_0^\pi \sin t dt \\ \int_0^\pi \cos t dt &= \sin t \Big|_0^\pi \\ &= \sin(\pi) - \sin(0) = 0 \\ \int_0^\pi \sin t dt &= -\cos t \Big|_0^\pi \\ &= -\cos(\pi) + \cos(0) = -(-1) + 1 = 2 \\ \int_0^\pi e^{it} dt &= 0 + i(2) = 2i\end{aligned}$$

Example 7

Evaluate $\int_0^{\pi/2} \cosh(it) dt$

Solution 7

$$\cosh(it) = \frac{e^{it} + e^{-it}}{2}$$

$$\text{since } e^{it} = \cos t + i \sin t.$$

$$\begin{aligned} \text{And also } e^{-it} &= \cos(-t) + i \sin(-t) \\ &= \cos t - i \sin t. \end{aligned}$$

$$\cosh(it) = \frac{(\cos t + i \sin t) + (\cos t - i \sin t)}{2}$$

$$\cosh(it) = \frac{2 \cos t}{2} = \cos t$$

Solution 7 CONT'D

$$\begin{aligned}\int_0^{\pi/2} \cosh(it) dt &= \int_0^{\pi/2} \cos t dx \\&= \sin t \Big|_0^{\pi/2} \\&= \sin\left(\frac{\pi}{2}\right) - \sin(0) \\&= 1 - 0 \\&= 1 + 0i\end{aligned}$$

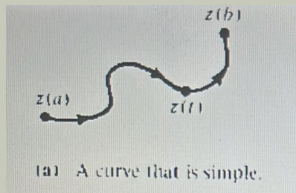
CONTOUR INTEGRAL

Definition: A Curve

A curve C in the plane is defined by parametric equations $C : z(t) = x(t) + iy(t)$ for $a \leq t \leq b$.

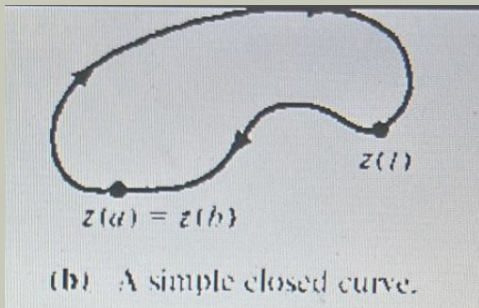
Definition: A Simple curve

A curve C is said to be simple if it does not cross itself, which is expressed by requiring that $z(t_1) \neq z(t_2)$ whenever $t_1 \neq t_2$.



Definition 3: A simple closed curve

A curve C is said closed if it has the property that $z(a) = z(b)$,

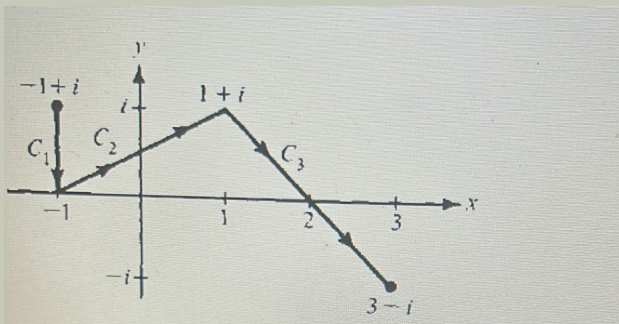


Definition : A Contour

Let $C_1, C_2, \dots, C_n$ denote n smooth curves such that the terminal point C_k coincide with the initial point of C_{k+1} for $k = 1, 2, 3, \dots, n$. Then the contour C is expressed by the equation $C = C_1 + C_2 + \dots + C_n$

Example 8

Find a parametrization of the polygonal path C from $-1 + i$ to $3 - i$ as shown by



The polygonal path $C = C_1 + C_2 + C_3$ from $-1 + i$ to $3 - i$

Solution to example 8

set $z_0 = -1 + i$ and $z_1 = -1$, then the segment C_1 joining z_0 to z_1 is given by

$$\begin{aligned}c_1 : z_1(t) &= z_0 + t(z_1 - z_0) \\&= (-1 + i) + t(-1 - (-1 + i)) \\&= (-1 + i) + t(-1 + 1 - i) \\&= (-1 + i) + t(-i) \\&= -1 + i(1 - t) \text{ for } 0 \leq t \leq 1\end{aligned}$$

Definition: CONTOUR INTEGRAL

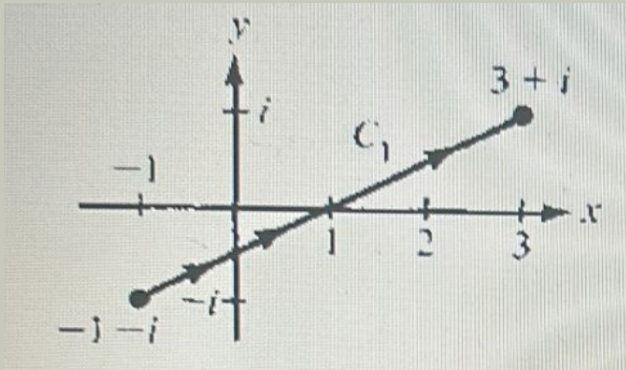
The contour integral of $f(z)$, a complex-valued function along C is defined to be

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

Example 9

Show that $\int_c z dz = 4 + 2i$ where c is the line segment from $-1 - i$ to $3 + i$.

Solution to Example 9



Solution to Example 9 CONT'D

The line segment joining $(-1, -1)$ to $(3, 1)$ is given by $y = \frac{1}{2}x - \frac{1}{2}$. This means that $x = 2y + 1$

Then let $y = t$ and $x = 2t + 1$

$$C : z(t) = (2t + 1) + it, \frac{dz}{dt} = (2 + i) \\ \Rightarrow dz = (2 + i)dt \text{ for } -1 \leq t \leq 1$$

$$\begin{aligned} \int_c z dz &= \int_{-1}^1 (2t + 1 + it)(2 + i) dt \\ &= \int_{-1}^1 (4t + 2it + 2 + i + 2it + i^2 t) dt \\ &= \int_{-1}^1 (4t + 4it + 2 + i - t) dt \\ &= \int_{-1}^1 (3t + 2) + i \int_{-1}^1 (4t + 1) dt \end{aligned}$$

Solution to example 9 CONT'D

$$\begin{aligned} &= \frac{3}{2}t^2 + 2t \Big|_{-1}^1 + i(2t^2 + t) \Big|_{-1}^1 \\ &= \frac{3}{2}(1)^2 + 2(1) - \left(\frac{3}{2}(-1)^2 + 2(-1) \right) + i(2(1)^2 + 1 - (2(-1)^2 + (-1))) \\ &= \left(\frac{3}{2} + 2 - \left(\frac{3}{2} - 2 \right) \right) + i(2 + 1 - (2 - 1)) \\ &= (3/6 + 2 - 8 + 2) + i(3 - 2 + 1) \\ &= 4 + 2i \end{aligned}$$

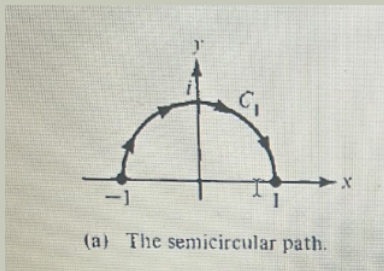
$$\begin{aligned} &= \frac{3}{2}(1)^2 + 2(1) - \left(\frac{3}{2}(-1)^2 + 2(-1) \right) + i(2(1)^2 + 1 - (2(-1)^2 + (-1))) \\ &= \left(\frac{3}{2} + 2 - \left(\frac{3}{2} - 2 \right) \right) + i(2 + 1 - (2 - 1)) \\ &= (3/6 + 2 - 8 + 2) + i(3 - 2 + 1) \\ &= 4 + 2i \end{aligned}$$

Example 10

Show that $\int_C \bar{z} dz = -\pi i$ where c is the semicircular path from -1 to 1.

Solution to Example 10

Consider the semi-circular path below



Solution to Example 10 continued

C can be parameterized by

$$C : z(t) = -\cos t + i \sin t$$

$$dz = (\sin t + i \cos t) dt$$

$$\begin{aligned} \int_0^\pi \bar{z} dz &= \int_0^\pi (-\cos t - i \sin t)(\sin t + i \cos t) dt \\ &= \int_0^\pi (-\sin t \cos t - i \cos^2 t - i \sin^2 t + \sin t \cos t) dt \\ &= -i \int_0^\pi (\cos^2 t + \sin^2 t) dt = -i \int_0^{(1)} dt = -i\pi \end{aligned}$$

Properties of Contour Integrals

If C is given by $C : z(t) = x(t) + iy(t)$ for $a \leq t \leq b$, then the contour integral for the opposite oriented contour $-C$ is given by

$$\int_{-C} f(z) dz = \int_{-b}^{-a} f(z(-\tau)) (-z'(-\tau)) d\tau$$

where $t = -\tau$

using a change of variables, we get

Properties of Contour Integrals

- $\int_{-C} f(z)dz = - \int_C f(z)dz.$
- $\int_C (f(z) \pm g(z))dz = \int_C f(z)dz \pm \int_C g(z)dz$
- $\int_C (c + id)f(z)dz = (c + id) \int_C f(z)dz$
- $\int_{C_1+C_2} f(z)dz = \int_{C_1} f(z)dz + \int_{C_2} f(z)dz$

THE END